

$$\frac{69}{99} = 69,7\%$$

STUDENT NAME AND SURNAME:

Jayden Wilson

VENUE:	Qualification number and Title:	
CHPC – Cape Town	118707 - Occupational Certificate: Software Developer	
SUBJECT:		
KM01 Informal Test: Computers and Computing Systems		
ASSESSMENT DATE:	18/02/2026	
SIGNATURE:		
STUDENT ASSESSMENT FEEDBACK:	It was a good test	
STUDENT FEEDBACK DATE:	18/02/2026	
ASSESSOR FEEDBACK:	Very good all-round knowledge displayed Jayden. I trust you will note and improve the topics which you missed in this assessment, as you continue to grow and master the knowledge, ahead of the formal assessment. Keep up the good work, and keep growing Wilson.	
ASSESSOR FEEDBACK DATE:	18/02/2026	
MODERATION DATE AND SIGNATURE		

MARKING SUMMARY			
QUESTION	TOTAL MARK	MARK	OUTCOME
Question 1			
Case Study	10	8	
Question total	10	8	
Question 2			
1.	5	4	
2.	5	4	
3.	6	6	
4.	5	—	
5.	4	4	
6.	4	4	
7.	4	4	
8.	2	1	
9.	4	—	
10.	3	—	
Question total	42	27	
Question 3			
1.	2	1	
2.	2	2	
3.	2	2	
4.	6	2	
5.	2	2	
Question total	14	9	
Question 4			
1.	4	2	
2.	4	4	
3.	1	1	
4.	2	1	
5.	2	2	
Question total	13	10	
Question 5			
1.	3	2	
2.	2	—	
3.	3	1	
4.	5	5	
5.	4	4	
7.	3	3	
Question-total	20	15	

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Final Mark	99	69	
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Informal Assessment

Instructions

- Candidates are required to complete the questions in full
- No internet access is permitted during the assessment.
- Close all browsers.
- Assessment duration is 150 minutes
- Student must pass with an aggregate of 80% to be declared competent
- Standard exam conduct and rules are to be maintained

Declaration

I.....Jayden Wilson..... (Learner) herewith declare that: I am ready for the assessment, that we have reviewed the assessment preparation and plan, I understand the assessment process and I am happy that the assessment will be conducted in a fair manner.

Learner signature:	Date: 18/02/2026
	
Assessor signature:	Date:
	18/02/2026

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Question 1

Case Study: The "Compile-Time Crash"

Scenario: Thabo is a junior developer at a tech firm in Johannesburg. He recently upgraded his workstation by adding a second high-end Graphics Card (GPU) to help with video rendering and machine learning tasks. However, ever since the upgrade, the computer behaves normally while he is writing code, but as soon as he starts a "Heavy Build/Compile" or runs a GPU-intensive simulation, the entire system randomly shuts down without any error message or "Blue Screen of Death."

Instructions: Using the 7 steps of the troubleshooting process, explain how Thabo should systematically resolve this issue to get his workstation back to a stable state. (10 Marks)

1. Gather information on the problem.

Note what I already know. This is:

- Thabo installed a second GPU into his computer.
- Ever since this installation, whenever he runs a Heavy Build/Compile, the entire system shuts down **WITHOUT** any error messages, or "Blue Screen of Death".

2. State the most probable cause.

Possible causes:

- The computer was not properly configured into handling or managing the new GPU properly.
- The GPU is simply too powerful for the current setup Thabo has, for example, his RAM may not be able to process the applications being run with the new GPU.
- Thabo could have affected the other components while installing his new GPU (unlikely).

Most probable cause:

- The configurations of his computer are not correctly set up to accommodate his new GPU, it may have trouble spreading resources between his old GPU and his new one.

3. Testing.

- Test my theory by installing a GPU such as Thabo's on my own computer, then opening my computers settings, Task Manager and whatever settings necessary in order to set up the distribution of resources within the computer. Perhaps even setting it up so that

whilst your technical accuracy still needs work
you made good use of the problem solving steps



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the new GPU must handle all the GPU heavy programs. Test this by running these programs and see how well it does.

4. Plan to implement the solution.

Notify Thabo that I will need time to test my theory on his PC. With a recommendation of doing this outside of his work hours, or during his break time. I will implement fallbacks in case of his device crashing such as a spare GPU or just resetting his settings to how they were originally.

5. Implement the solution.

8 Marks

Take Thabo's PC and configure his settings accordingly.

6. Verify functionality and implement preventative measures.

Verify whether my solution works on Thabo's PC. Implement preventative measures by notifying him to adjust his configurations whenever installing new components on his PC. I will create a formal document explaining to the rest of the company on how to check if their current device is configured correctly, and how to change their configurations accordingly.

7. Document Findings.

Problem: Thabo installed a second and improved GPU into his PC and ever since, it started crashing whenever running GPU heavy programs.

Cause: Thabo's PC was not configured to properly manage resources between his old GPU and his new one and therefore was crashing consistently.

Solution: Open up the PC **Task manager > Resource Management > Resource > GPU & graphics**. Configure resources accordingly, with the new components in mind.

Time of completion: It took 15 minutes to configure the PC settings.

Date & Time of problem: 18 February 2026, Thabo called at 9:00 AM. I had it fixed during his break time.

Question 2

1. List five common hardware symptoms that would indicate a failing Power Supply Unit (PSU) before it stops working entirely. (5 Marks)

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1. The supply unit emits sparks whenever it is plugged in to a power source, such as an outlet. 2. The power unit overheats in a short amount of time. 3. The PC charges slowly. 4. PC does not charge and does not power on due to low battery. 5. The cables used with the power supply unit are exposed.
2. Describe the precautions a technician must take regarding Static Electricity (ESD) when opening a computer case to install internal components. (5 Marks)
- Remove any jewelry, to ensure that no electricity travels to your body - Wear an anti-static wrist strap, this serves as an insulator. - Switch off the power at the Power source, then disconnect the power source. - Ensure that the components are fully switched off.
3. Name and briefly describe the function of three different ports found on a modern motherboard's I/O panel (6 Marks)
- USB Port – This is a means to transfer data to and from the PC. - HDMI/ VGA Port – This is a means to display and/or transfer video and audio. - Audio Port – Transfer audio.
4. Explain the role of the Front Side Bus (FSB) or its modern equivalents (like DMI) in facilitating communication between the CPU and the motherboard. (5 Marks)
X
5. Differentiate between 32-bit and 64-bit processors in terms of how much RAM they can effectively address. (4 Marks)
A 32-bit processor is able to address up to 4 gbs of ram. The 64-bit processor is the current standard. It can process up to 16gbs of RAM and is much faster and more efficient than the 32 bit processor.
6. Describe the difference between Cache memory (L1/L2/L3) and standard System RAM in terms of speed and proximity to the CPU. (4 Marks)
Cache memory is located closer to the CPU than RAM

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Cache memory is faster than RAM ✓ ✓
7. What is UEFI, and how does it differ from the traditional BIOS found in older systems? (4 Marks)
UEFI is a modern version of the BIOS System, it has better GUI and more options to work with when compared to BIOS. It is faster and more user friendly. ✓ ✓
8. What are the common symptoms a user might see if the CMOS battery on the motherboard has failed? Type text here (2 Marks)
They will notice that the clock on their PC is inaccurate. The date on the PC will also be inaccurate. ✓
9. Compare NVMe M.2 SSDs to traditional SATA SSDs in terms of physical form factor and data transfer speeds. (4 Marks)
✗
10. With the aid of a brief explanation, why is the 80 PLUS rating important when selecting a power supply for a high-performance workstation? (3 Marks)
So that you can assign the correct power supply to the working station. The 80 PLUS rating is an important indicator of what a Power supply's specs are. This is the standard with all power supplies. ✗

Question 3

1. Differentiate between a Personal Area Network (PAN) and a Metropolitan Area Network (MAN). (2 Marks)
A PAN covers a local area such as a house or a building. While a MAN covers a wider area such as an entire city. ✓
2. What is the primary purpose of a Network Interface Card (NIC), and can a computer function on a network without one? (2 Marks)
A Network Interface Card is necessary to accessing the internet and connecting to a network. A computer will not be able to function on a standard network without a Network Interface card, as they will need to be connected to the network. ✓
3. Describe the effect of disabling SSID broadcasting on a wireless router from a security perspective. (2 Marks)

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This keeps the wireless router more secure as now the SSID will not be broadcasted publicly. It is now private and will be shared with permission. ✓
4. Explain the functional difference between a Public IP address and a Private IP address. (6 Marks)
A public IP address can be accessed publicly, anybody in the world can see your public IP Address, while nobody else can see your Private IP address ✓
5. Identify two risks associated with using public unencrypted WiFi when accessing cloud collaboration services. (2 Marks)
There may be hackers on the public Wi-Fi network and they will now have access to your cloud services. They can steal your information. ✓

Question 4

1. Classify the following as Input (I), Output (O), or Both (B) and explain their function: (a.) Webcam (b.) External Hard Drive. (4 Marks)
Webcam – Input: This is used to capture a live video or to record a video of a person, it is then displayed on a monitor or PC screen. ✓ External Hard Drive – Input: This is actually a storage device, but it can also be classified as an input device since it is used to give information to devices. An External Hard drive does receive information, but it only stores it. It does not display information and therefore is not an output device. ✗
2. A user reports their USB mouse is not responding. List two troubleshooting steps to determine if the issue is with the mouse or the computer port. (4 Marks)
1. Test the computer port with a different USB, such as a storage device and see if it connects. ✓ 2. Test the mouse on a different computers USB port and see if it connects successfully. ✓
3. Name one hardware component found in a smartphone that is usually absent in a standard desktop (1 Mark)
Touchscreen. ✓
4. List two preventative maintenance steps specifically for extending the battery life of a laptop or mobile device. (2 Marks)
• Do not keep it connected to a charger overnight. ✗

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• Add a charge blocker – This is to limit how much a battery can charge to, for example, A phone will stop charging once it reaches 90%. This causes less strain on the phone's battery, therefore extending its life.
5. For each of the following tools, explain the primary function: a.)Disk Cleanup Utility b.)Multimeter (used by a hardware technician). (2 Marks)
Disk Cleanup Utility: This is a software Tool used to inspect storage on a PC, and to clear memory. Multimeter: A hardware tool used to measure electricity.

Question 5

1. Explain how the Operating System manages Virtual Memory when the physical RAM is fully utilized. (3 Marks)
The Operating System begins distributing the remaining memory using Virtual Memory, this is much slower than RAM
2. What is the general purpose of the NTFS file system compared to the older FAT32 system? (2 Marks)
X
3. Describe the steps to Roll Back a device driver in Windows if a new update causes system instability. (3 Marks)
Have a back up available.
4. Compare a System Restore to a Factory Reset in terms of what happens to the user's personal files. (5 Marks)
With a factory reset, all the memory, storage, applications, data on a device is wiped. Only the necessary programs that comes with the device remains. It is as if the device is brand new and like nobody used it before. A System Restore only resets the necessary systems on a device. The users information is barely affected.
5. Justify the importance of Encryption (such as BitLocker) for protecting data on a lost or stolen laptop. (4 Marks)
Encryption will make data unreadable and inaccessible. This is important when a device is lost or stolen because whoever has the device cannot make sense of the data available. They will need

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the proper decryption tool to access the encrypted information. Any lost data will at least be protected until it is recovered.

6. List three distinct types of Cloud Service Models that a software developer might use. (3 Marks)

SaaS

IaaS

PaaS

6

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